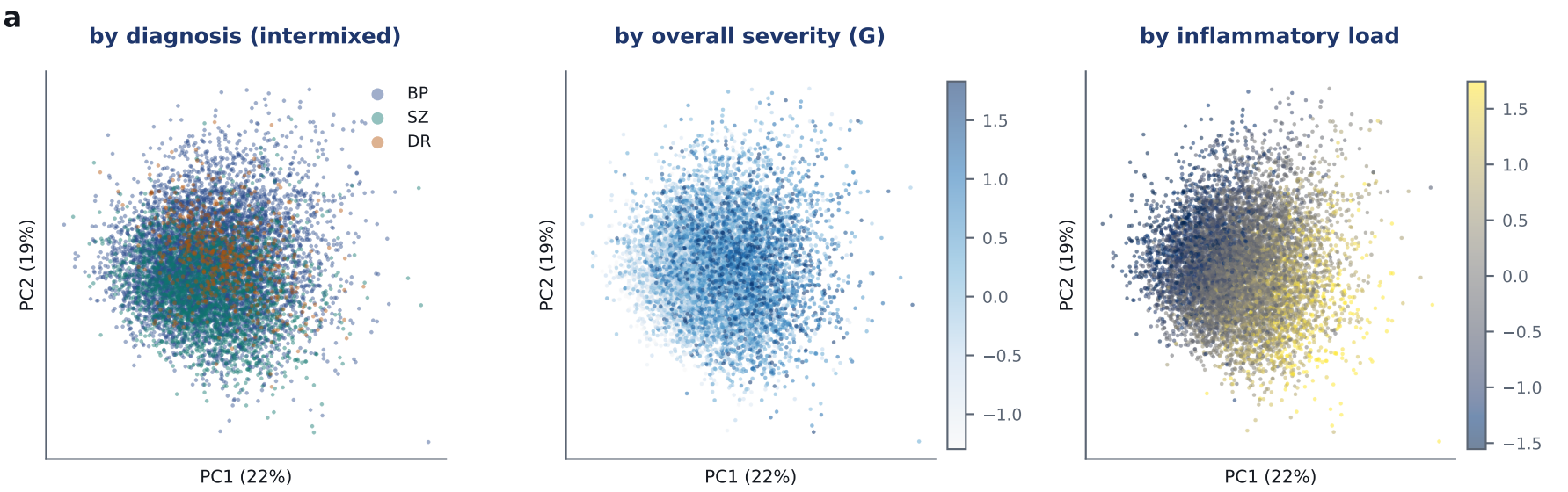




b **Structure gate → continuum**

Gap statistic	K = 1
HDBSCAN	0 clusters
Hartigan dip (PC1)	p ≈ 0.99
Archetype scree	no elbow
Mixture BIC	flat basin

no discrete biotypes

